

```
#include <systemc.h>

// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;

    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;

    void consume() {
        int value;
        while (true) {
```

[Wrote 50 lines]

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```
praveenkumar@gmail:~$ nano Exp25.cpp
praveenkumar@gmail:~$ g++ -o Exp25 Exp25.cpp -lsystemc
praveenkumar@gmail:~$ ./Exp25
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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```

```
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
```

```
praveenkumar@gmail:~$
```